

Introducing MySQL

In this chapter, you'll learn what MySQL is, and you'll learn about the tools you can use when working with it.

What Is MySQL?

In the previous chapter, you learned about databases and SQL. As explained, it is the database software (the *DBMS* or *database management system*) that actually does all the work of storing, retrieving, managing, and manipulating data. MySQL is a DBMS; that is, it is database software.

MySQL has been around for a long time, and it is now installed and in use at millions of installations worldwide. Why do so many organizations and developers use MySQL? Here are some of the reasons:

- **Cost:** MySQL is open source, and it is usually free to use and even modify the software.
- **Performance:** MySQL is fast—make that very fast.
- **Trustworthiness:** MySQL is used by some of the most important and prestigious organizations and sites, all of which entrust it with their critical data.
- **Simplicity:** MySQL is easy to install and get up and running.

In fact, historically the only real criticism of MySQL has been that it has not always supported the functionality and features offered by other DBMSs. But as new features have been added to each new version, that has changed, and it continues to do so.

Client/Server Software

DBMSs fall into two categories: shared file-based and client/server DBMSs. Shared file-based DBMSs (which include products such as Microsoft Access and FileMaker) are designed for desktop use and are generally not intended for use on higher-end or more critical applications.

Databases such as MySQL, Oracle, and Microsoft SQL Server are client/server-based databases. Client/server applications are split into two distinct parts. The *server* portion is a piece of software that is responsible for all data access and manipulation. This software runs on a computer called the *database server*.

Only the server software interacts with the data files. All requests for data, data additions and deletions, and data updates are funneled through the server software. These requests or changes come from computers running client software. The *client* is the piece of software with which the user interacts. If you request an alphabetical list of products, for example, the client software submits that request over the network to the server software. The server software processes the request; filters, discards, and sorts data as necessary; and sends the results back to your client software.

Note

How Many Computers? The client and server software may be installed on two computers or on one computer. Either way, the client software communicates with the server software for all database interaction.

All this action occurs transparently to you, the user. The fact that data is stored elsewhere, or that a database server is performing all this processing for you, is hidden. You never need to access the data files directly. In fact, most networks are set up so that users have no access to the data or even the drives on which it is stored.

Why is this significant? Because to work with MySQL, you'll need access to both a computer running the MySQL server software and client software with which to issue commands to MySQL:

- The server software is the MySQL DBMS. You can be running a locally installed copy, or you can connect to a copy running on a remote server to which you have access.
- The client can be MySQL-provided tools (more on these below), scripting languages (such as Python and Ruby), web application development languages and platforms (such as ASP.NET, JavaScript, and Node.js), programming languages (such as C, C++, and Java), and more.

Note

Cloud-Based DBMSs You may have come across cloud-based DBMSs, which are hosted database services that are accessed from a web browser. These are technically client/server DBMSs; the client just happens to be code that runs in the browser.

MySQL Versions

We'll get back to client tools in a moment. First, a quick word about DBMS versions.

The current version of MySQL is version 8. Version 5 (all the way through 5.7) is still in use in many organizations. (Version 6 was never fully released, and there was no version 7.)

This book was written with MySQL version 8 in mind, but most of the chapters apply to version 5, too.

Note

Version Requirements Noted Any chapter that requires a specific version of MySQL clearly notes that at the start of the chapter.

MySQL Tools

As just explained, MySQL is a client/server DBMS, and so to use MySQL, you'll need a client—an application that you use to interact with MySQL (by giving it commands to be executed).

There are lots of client application options, but when learning MySQL (and, indeed, when writing and testing MySQL scripts), you are best off using a utility designed for just that purpose. And there are two tools in particular that warrant specific mention: the `mysql` command-line utility and the interactive MySQL Workbench.

`mysql` Command-Line Utility

Every MySQL installation comes with a simple command-line utility called `mysql`. This utility does not have any drop-down menus, fancy user interfaces, mouse support, or anything like that.

Typing `mysql` at your operating system command prompt brings up a simple prompt that looks like this:

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.31 MySQL Community Server - GPL
Copyright (c) 2000, 2022, Oracle and/or its affiliates.
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

Note

MySQL Options and Parameters If you just type `mysql` by itself, you might receive an error message. If you do, it is likely because security credentials are needed or because MySQL is not running locally or on the default port. `mysql` accepts an array of command-line parameters you can (and might need to) use. For example, to specify the user login name `ben`, you use `mysql -u ben`. To specify a username, host name, and port and get a prompt for a password, you use `mysql -u ben -p -h myserver -P 9999`.

You can obtain a complete list of command-line options and parameters by using `mysql --help`.

Of course, your version and connection information might differ, but you will be able to use this utility anyway. You'll note that:

- Commands are typed after the `mysql>` prompt.
- Commands end with `;` or `\g`; in other words, just pressing Enter will not execute a command.
- Type `help` or `\h` to obtain help. You can also provide additional text to obtain help on specific commands (for example, `help select` to obtain help on using the `SELECT` statement).
- Type `quit` or `exit` to quit the command-line utility.

The `mysql` command-line utility is one of the most used utilities and is invaluable for quickly testing and executing scripts. In fact, all of the output examples used in this book were created with `mysql` command-line output.

Tip

Familiarize Yourself with the `mysql` Command-Line Utility Even if you opt to use the graphical tool described next, you should make sure to familiarize yourself with the `mysql` command-line utility, as this is the one client you can safely rely on to always be present (as it is part of the core MySQL installation).

MySQL Workbench

MySQL Workbench is a graphical interactive client used to write and execute MySQL commands.

MySQL Workbench consolidates and replaces several different interactive tools used in prior versions of MySQL. Since its release, this tool has quickly become a developer favorite.

Note

Obtaining MySQL Workbench Unlike the command-line utility `mysql`, MySQL Workbench is not always installed as part of the core MySQL DBMS installation. If it isn't, you can download it for free directly from <https://dev.mysql.com/downloads/workbench/>. (Versions are available for Linux, macOS, and Windows, and source code is downloadable, too.)

We'll start using MySQL Workbench in the next chapter as we explore connecting to and using the MySQL DBMS. For now, note the following:

- MySQL Workbench features a color-coded editor that can help you write SQL statements.
- You can test your SQL statements right inside MySQL Workbench, and results (if there are any) will be displayed in a grid right below your statements.

- MySQL Workbench also lists all available datasources (called schemas). You can expand any datasource to see its tables, and you can expand any table to see its columns.
- You can use MySQL Workbench to edit data, check the server status and settings, back up and restore data, and much more.

Tip

Execute Saved Scripts You can use MySQL Workbench to execute saved scripts. To do this, select File, Open SQL Script, select the script (which will be displayed in a new tab), and click the Execute button (the one with a yellow lightning bolt).

MySQL Workbench is an important tool, and I suggest that you use it for all chapters in this book. We'll look at it in detail in the next chapter.

Other Tools

You are not limited to using MySQL's provided clients. There are lots of third-party clients out there, too, and you can use any of them with MySQL. These are some popular ones:

- DBeaver
- HeidiSQL
- phpMyAdmin
- RazorSQL

To use any of these tools, you generally just need to provide server and login details.

Summary

In this chapter, you learned what exactly MySQL is. You were also introduced to two client utilities: an included command-line utility and an optional but highly recommended graphical utility.